ENRIQUE GARCIA RIVERA

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EDUCATION

Washington University in St. Louis St. Louis, MO
Bachelor of Science in Mechanical Engineering | GPA: 3.65 / 4.00 Expected May 2027
Relevant Coursework: Flight Dynamics and Aircraft Design (graduate, in progress); Modeling, Simulation, and Control; Numerical Methods and Matrix Algebra; Vibrations; Fluid Mechanics

RESEARCH EXPERIENCE

UAS Research Intern, Missouri Institute for Defense and Energy, Kansas City, MO Summer 2026

- Built a MATLAB pipeline that batch-processes ArduPilot flight logs from 3 tactical fixed-wing Group 2 UAS, automatically detecting 125 flights in one month and organizing flight activity by aircraft and day.
- Developed an interactive MATLAB center of gravity model from measured landing-gear weights to evaluate payload layouts and solve for the nose ballast needed to hold the design CG.
- Rebuilt quadcopters from component-level hardware for a field-experimentation fleet, configuring ArduPilot over MAVLink and DroneCAN with per-vehicle parameter baselines and calibration.
- Supported UAS flight testing in the field, including ground station setup and composite airframe repair.

Research Intern, Multiplatform Interactive Robotics Laboratory, Kansas City, MO Summer 2025

- Built a Teensy 4.0 embedded sensing platform synchronizing two EMG and two IMU streams at 1 kHz, and ran timed trials across rest, contraction, and motion conditions.
- Processed EMG and IMU signals in Python (filtering, RMS envelopes, FFT); contributed to an IEEE Body Sensor Networks 2025 abstract, Toward Electromechanical Measurement of Muscle Activity.

ENGINEERING EXPERIENCE

Founding Member and Lead Avionics Engineer, WashU VTOL, St. Louis, MO Sep. 2025 – Present

- Lead avionics and flight control integration for the team's Vertical Flight Society Design-Build-Vertical Flight aircraft: a hexacopter prototype (2025–26) and the current 2026–27 quadcopter.
- Integrated the hexacopter's flight control and sensing system around a Pixhawk 6C running PX4 with isolated 4S/6S power; configured flight modes and failsafes and tuned PID gains.
- Traced barometer and magnetometer dropouts to motor-induced electromagnetic interference on the I2C bus and restored reliable sensor data by rerouting lines, twisting signal pairs, and adding pull-ups.
- Validated the integrated avionics in stages: bench, hardware-in-the-loop simulation, and first hover.

Structures and Test Engineer, WashU Design/Build/Fly, St. Louis, MO Aug. 2024 – Present

- Sized primary wing structure for AIAA Design/Build/Fly and SAE Aero Design aircraft under 2.5 g loads using ANSYS finite element analysis, iterating on carbon-fiber spar layup.
- Designed and fabricated mission payload systems, including a fuel-carrying composite pylon and a winch deployment system, and built a drop-test stand to verify payload protection.

TEACHING

Undergraduate Teaching Fellow, Washington University in St. Louis Aug. 2025 – Present

- Hold office hours and grade for Structural Dynamics and Vibrations (Fall 2026) and Electrical and Electronic Circuits (Fall 2025), helping students connect theory to physical systems.

TECHNICAL SKILLS

Dynamics, Control, and Analysis: MATLAB/Simulink, Python, C++, dynamics modeling, feedback control, PID tuning, signal processing, ANSYS FEA, SolidWorks
Flight Systems and Test: PX4, ArduPilot, MAVLink, DroneCAN, Pixhawk, HIL testing, embedded systems, data acquisition, sensor calibration, composites